ESP32-C3 Board in KiCad

1 Project Setup

- New KiCad Project
- Name → "esp32" in its own folder
- Open schematic

2 Schematic

2.1 ESP32 Module

- "a" → add `esp32-c3-wroom-02` (onboard antenna)

2.2 USB-C

- Add: `USB_C_Receptacle_USB2.0_16P`
- With footprint: `HRO_TYPE-C-31-M-12`
- Add GND connection to GND and Shield
- Add 5.1k pull-down resistors from CC1 and CC2 to GND
- All resistors, capacitors and the LED use 0603 footprints:
  - `Resistor_SMD:R_0603_1608Metric`
  - `Capacitor_SMD:C_0603_1608Metric`
  - `LED_SMD:LED_0603_1608Metric`

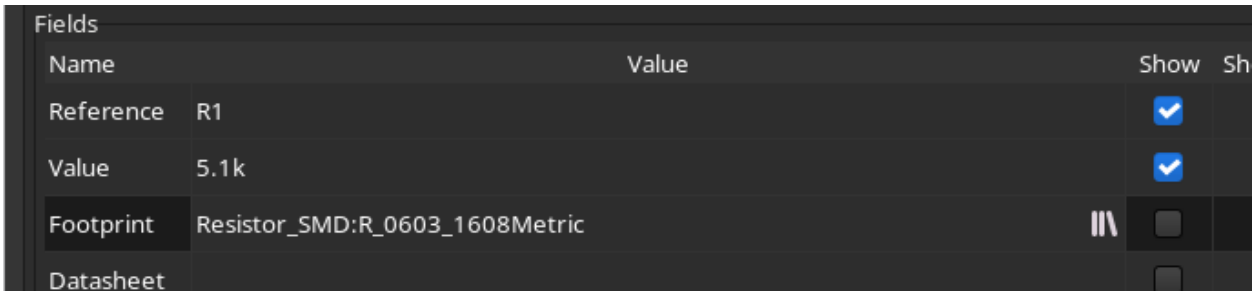


Figure 1: Fields for R1.

- "Q" no connect SBU1 and SBU2

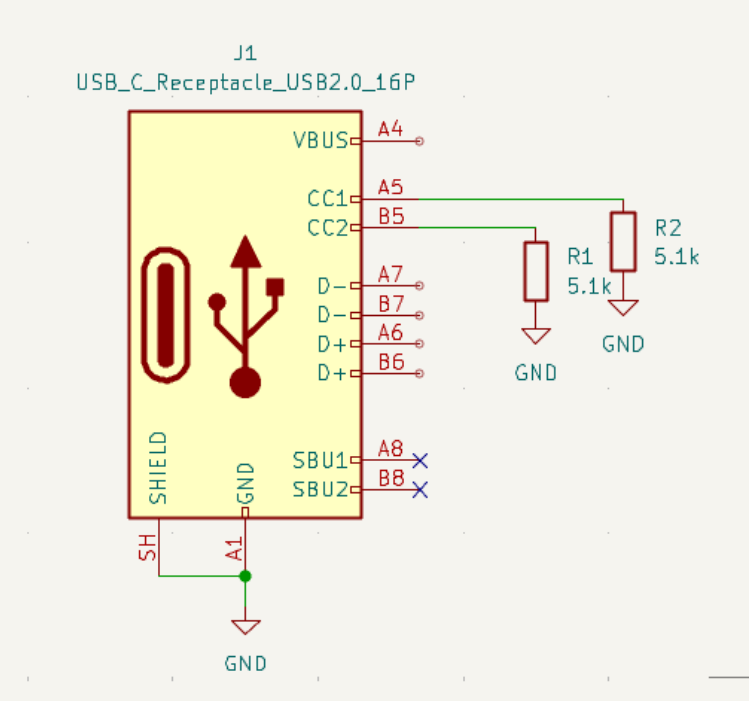


Figure 2: USB-C receptacle with CC pull-downs and SBU no-connects.

- +5V flag at VBUS
- PWR flag on +5V and GND

2.3 Regulator

`AP2112K-3.3`

- VIN: +5V
- Decoupling cap 10 µF rated at 10 V to GND
- Tie EN to VIN
- GND to GND
- Decoupling cap 10 µF rated at 10 V to VOUT
- VOUT to 3.3V flag

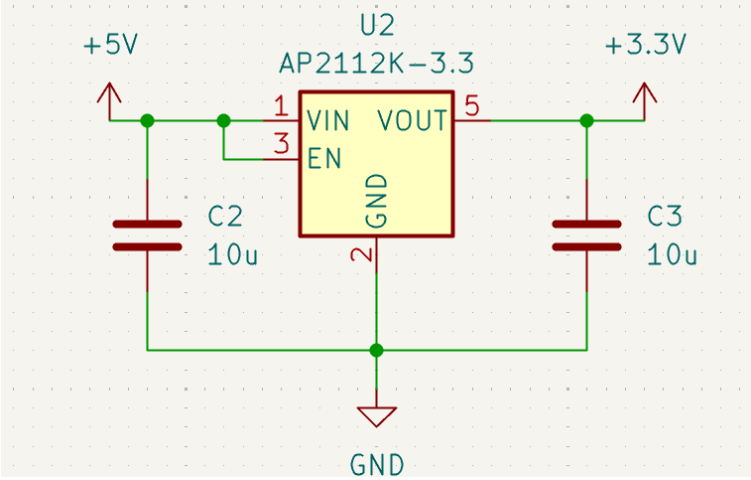


Figure 3: AP2112K-3.3 regulator with input and output decoupling.

2.4 USB Data Lines

Use “L” to place a Net Label.

- D− tied together
- D− is USB−
- D+ tied together
- D+ is USB+

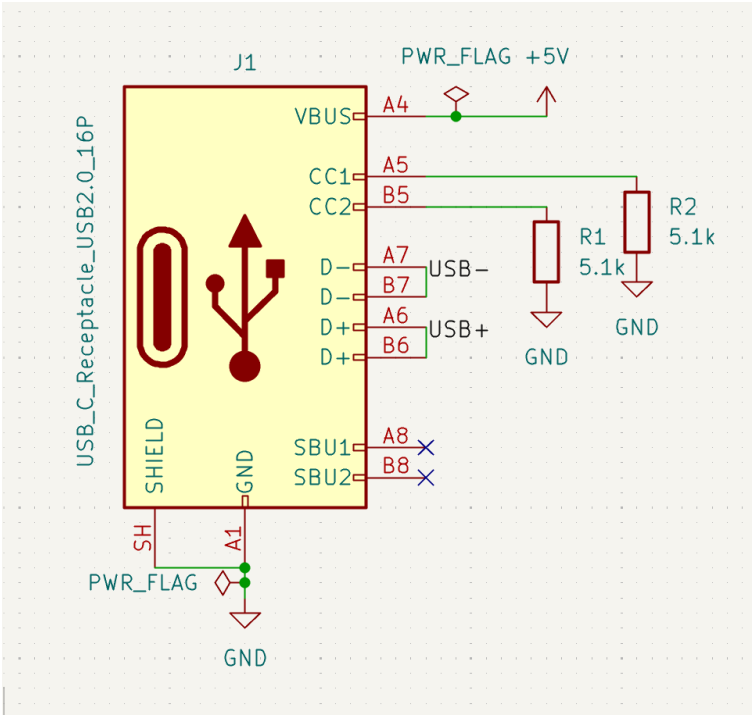


Figure 4: Completed USB-C connector.

2.5 Button

- Button: `SW_Push`
- Button footprint: `Button_Switch_SMD:SW_SPST_PTS810`

2.6 ESP32 Decoupling

- 3.3V to 3V3
- Add 0.1 µF to GND, 16 V rated
- Add 10 µF to GND, 16 V too
- Pull up 10k from EN to 3.3V
- Add decoupling cap 1 µF to EN
- IO9: pull up 10k to 3.3V
- Add switch to GND

2.7 I2C

- 4.7k pull-up to 3.3V on pins 2 and 8
- SDA: IO2
- SCL: IO8

2.8 Status LED

- IO10 → 1 kΩ resistor → 0603 LED → GND

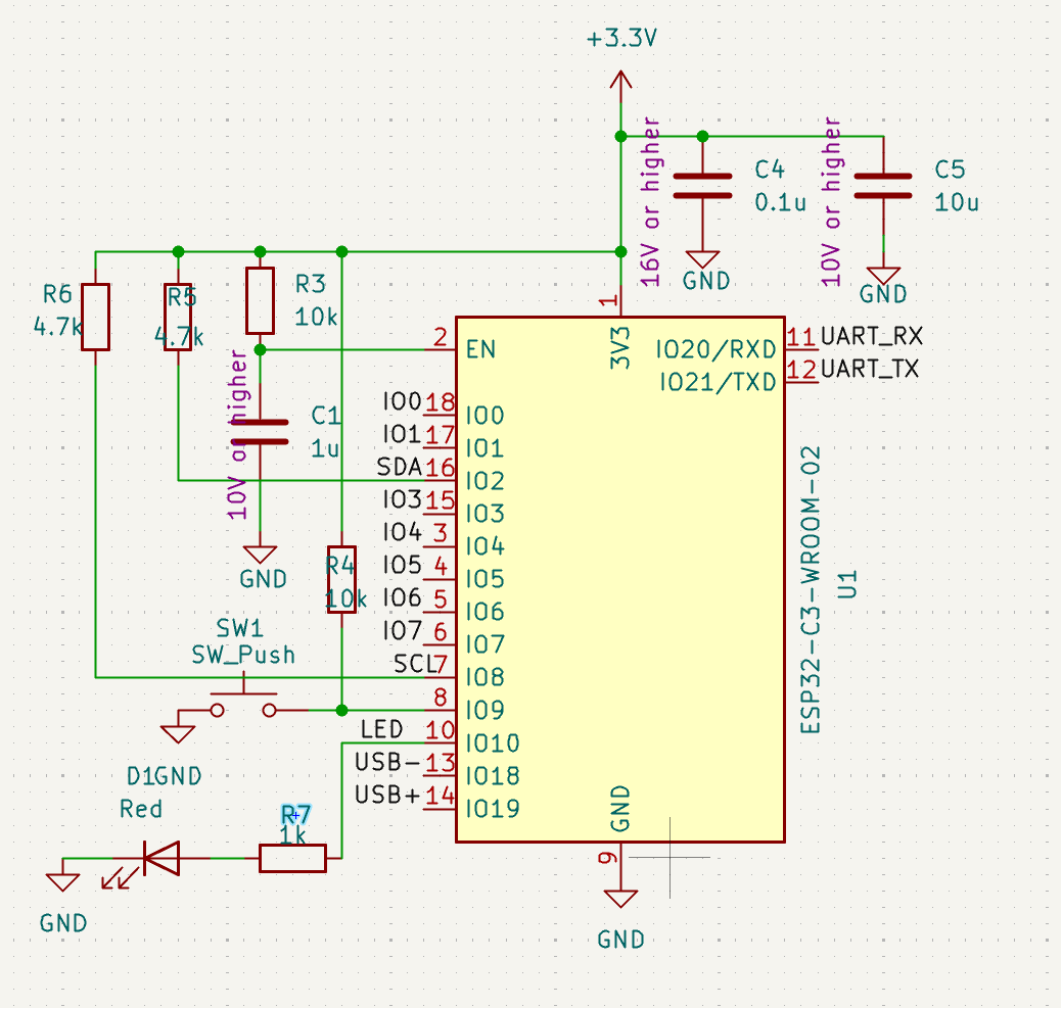


Figure 5: ESP32-C3 module with decoupling, EN, boot button, I2C pull-ups and status LED.

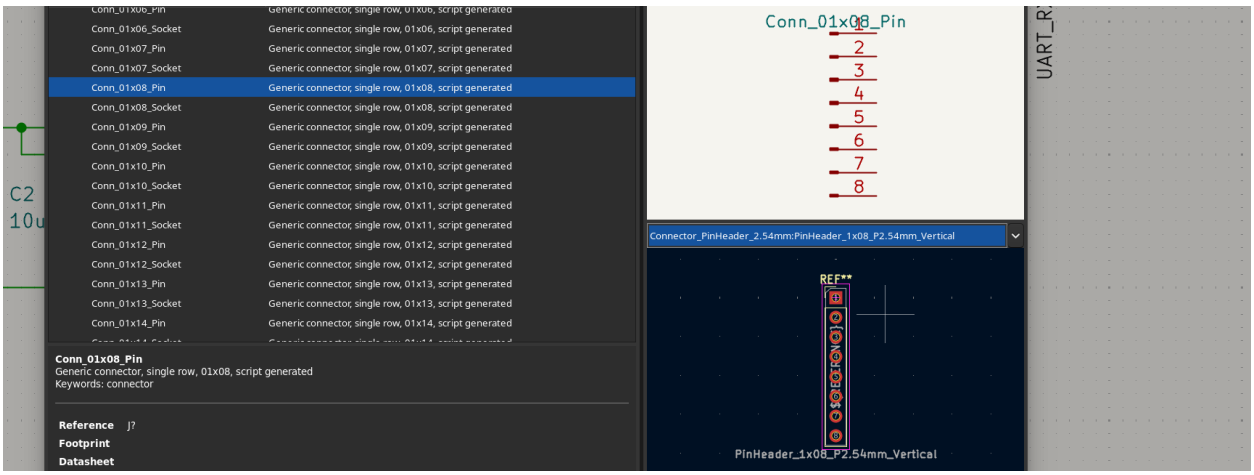


Figure 6: `Conn_01x08_Pin` with footprint `PinHeader_1x08_P2.54mm_Vertical`.

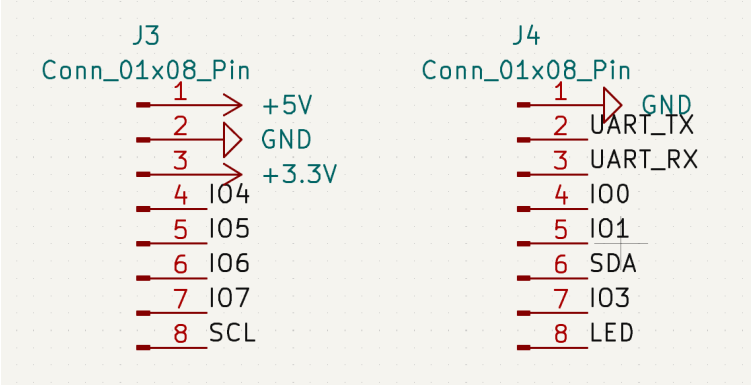


Figure 7: J3 and J4 pin headers.

- J3 (left side of U1): +5V, GND, +3.3V, IO4, IO5, IO6, IO7, SCL
- J4 (right side of U1): GND, UART_TX, UART_RX, IO0, IO1, SDA, IO3, LED

2.9 Checking the Schematic

1. **ERC:** Inspect → Electrical Rules Checker → Run ERC. Aim for zero errors, then read each warning instead of ignoring it.
2. **Highlight a net:** select a wire and press ` (backtick) to highlight everything on that net. Click SDA and SCL in turn and make sure each lights up only its own pins.

3 PCB Editor

3.1 Board Setup

1. Update PCB from schematic (F8)
2. Change grid to 0.25 mm
3. Click the green PCB icon with the gear (Board Setup)
4. Constraints: minimum clearance 0.15 mm, minimum track width 0.15 mm, minimum via diameter 0.6 mm, copper to edge 0.3 mm (Figure 8)
5. Net Classes: add a `Power` class with 0.5 mm track width. Leave `Default` at 0.25 mm tracks, 0.6/0.3 mm vias, 0.2 mm differential pair width and 0.25 mm gap (Figure 9)
6. Netclass Assignments: set `+5V`, `+3.3V` and `GND` to **Power** (the screenshot still shows Default)
7. Pre-defined Sizes: tracks 0.25 mm and 0.5 mm, via 0.6 mm with 0.3 mm hole (Figure 10)

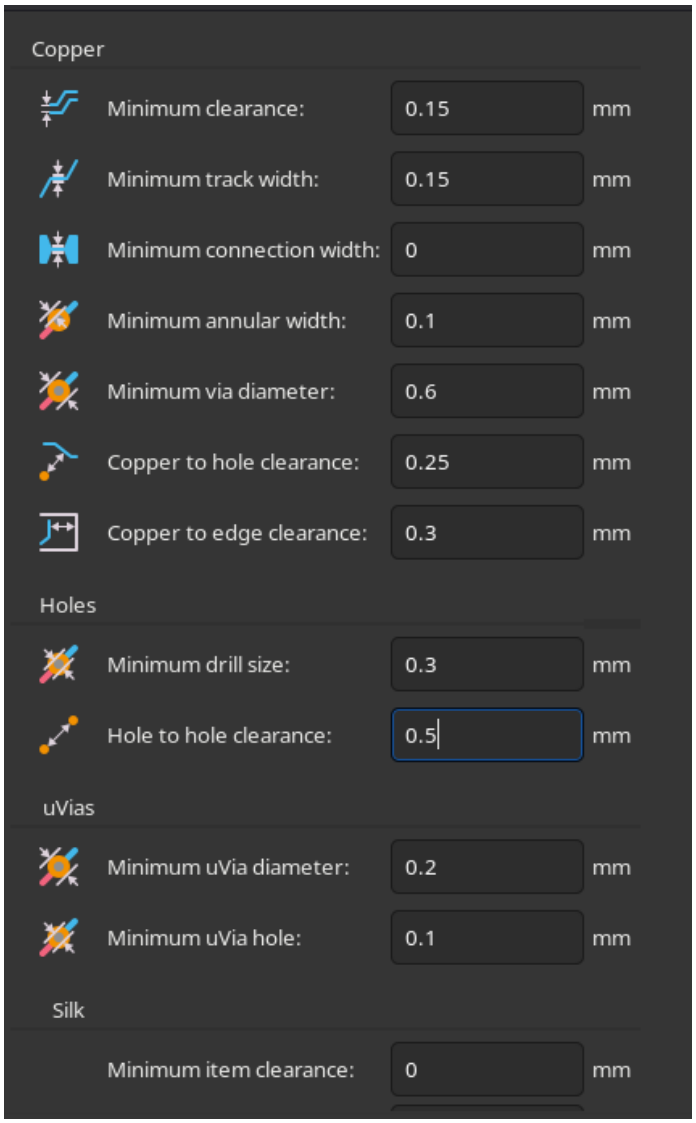


Figure 8: Board Setup → Design Rules → Constraints.

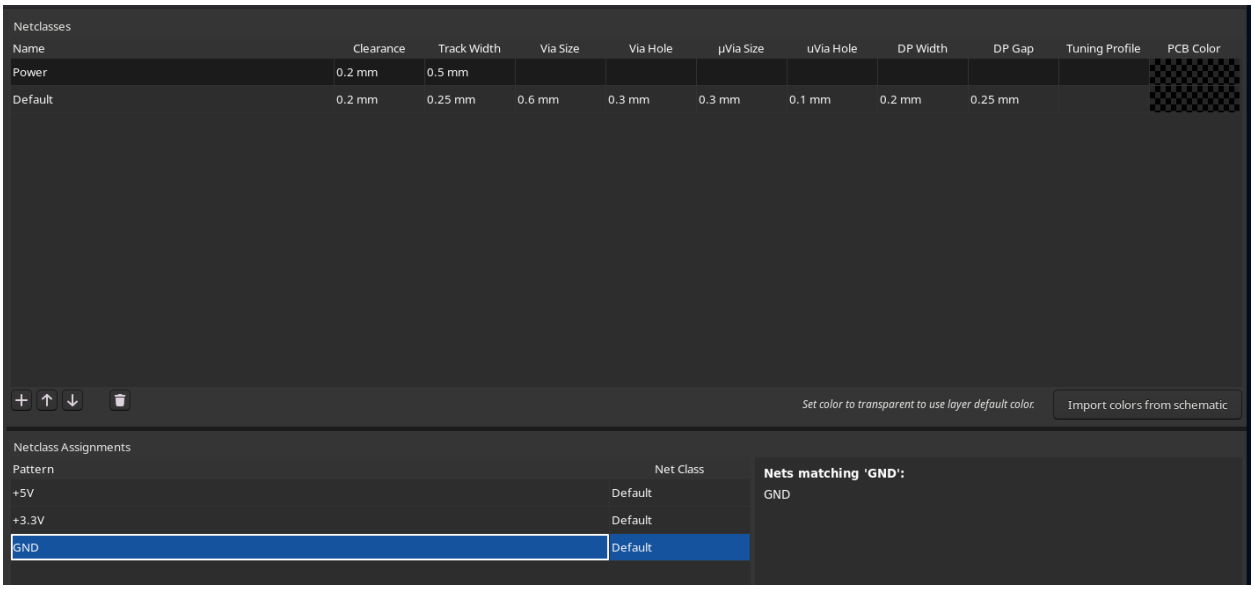


Figure 9: Board Setup → Design Rules → Net Classes.

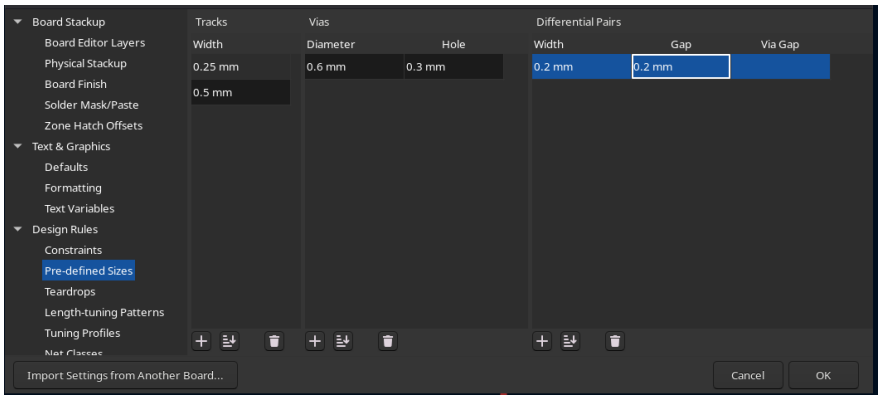


Figure 10: Board Setup → Design Rules → Pre-defined Sizes.

3.2 Edge cuts

1. Select Edge.Cuts under Appearance on the right side
2. Draw Rectangles
3. Double-click the rectangle and set size to 50x40 mm
4. Ctrl + Shift + M is the measure tool

3.3 Placement

1. Place U1 with the antenna hanging over the top board edge
2. Keep the antenna keep-out clear of copper and parts on both layers
3. Place J3 and J4 on a 2.54 mm grid along the left and right edges
4. Place J1 at the bottom edge so a cable can reach it
5. Lock U1, J1, J3 and J4 (select, then L)
6. C4 and C5 right next to U1 pin 1 (3V3)

7. C1 next to U1 pin 2 (EN)
8. C2 next to U2 VIN, C3 next to U2 VOUT
9. R1 and R2 next to the CC pins on J1
10. R3 to R6 between the headers and U1
11. R7 and D1 near the bottom edge so the LED is visible
12. SW1 near the bottom edge so it is easy to press

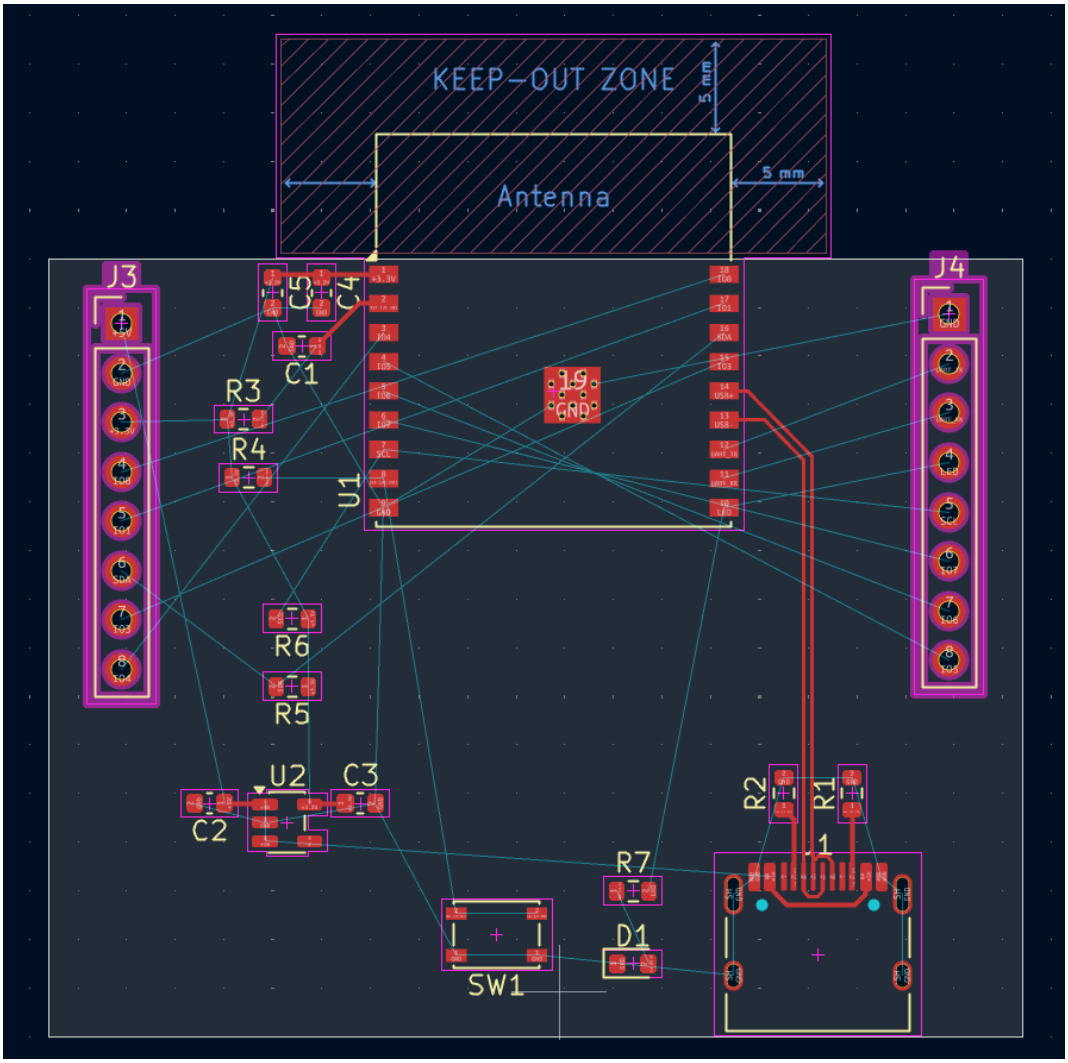


Figure 11: Placement with the USB pair routed.

3.4 Differential Router

- Press 6
- Click a USB+ or USB− pad on J1
- Route to U1 pins 13 and 14 on F.Cu
- No vias and nothing crossing the pair
- Join A6 to B6 and A7 to B7 at the connector

3.5 Routing

1. Press X to route a single track; the width follows the net class
2. Route power (+5V, +3.3V) first: J1 VBUS → C2 → U2 VIN, then U2 VOUT → C3 → C4/C5 → U1 pin 1
3. Then EN, IO9, I2C, the LED and the header pins
4. Keep tracks on F.Cu where possible; use a via (V while routing) only to hop over another track
5. Leave GND pins unrouted; the pour and stitching vias connect them
6. Nothing routed under the antenna

3.6 Ground Pour

1. Select B.Cu under Appearance on the right side
2. Place → Add Filled Zone (Ctrl + Shift + Z)
3. Click the first corner of the board outline
4. In the zone dialog:
   - Check B.Cu and F.Cu
   - Net: GND
   - Clearance: 0.5 mm
   - Minimum width: 0.25 mm
   - Pad connections: Thermal reliefs
   - Thermal relief gap: 0.5 mm
   - Thermal spoke width: 0.5 mm
   - Corner smoothing: None
   - Remove islands: Always
5. Click OK
6. Click the other corners of the board outline
7. Double-click to close the outline
8. Press B to fill all zones
9. Check that no copper fills the antenna keep-out

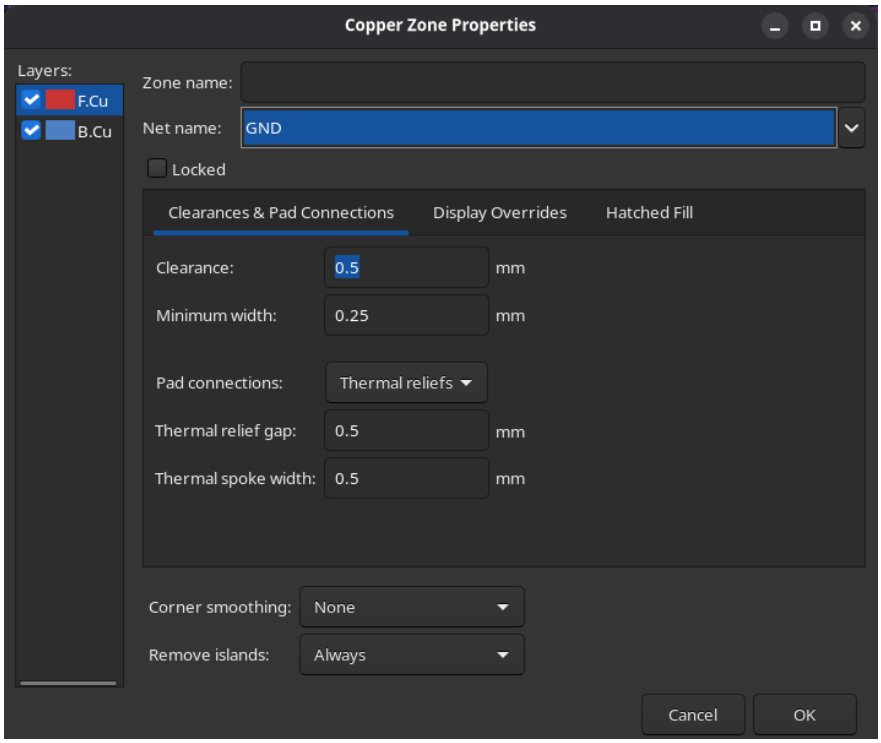


Figure 12: Copper Zone Properties for the GND pour.

3.7 Stitching Vias

1. Place → Add Vias (Ctrl + Shift + X)
2. Click on the GND pour; the via takes the GND net
3. One via every 5 to 10 mm across the board
4. Extra vias next to each capacitor GND pad, J1 and U2
5. 4 to 9 vias in the U1 GND pad (pin 19)
6. Press B to refill, then run DRC

3.8 Design Rules Check

1. Press B to refill zones
2. Inspect → Design Rules Checker → Run DRC
3. Check “Test for parity between PCB and schematic”
4. Fix every error; zero unconnected items
5. Click an entry to jump to it on the board
6. Re-run until clean

3.9 Silkscreen

1. Move reference designators so none sit on pads or under parts
2. Label the header pins (+5V, GND, SDA, . . .) with Place → Add Text on F.SilkS
3. Add a board name and date
4. View → 3D Viewer (Alt + 3) to check the result

3.10 Fabrication Outputs

1. File → Fabrication Outputs → Gerbers
2. Output directory: `gerbers/`
3. Layers: F.Cu, B.Cu, F.Paste, F.SilkS, B.SilkS, F.Mask, B.Mask, Edge.Cuts
4. Click Plot
5. Click Generate Drill Files → Generate (same folder)
6. Zip the `gerbers/` folder and upload it to the board house; check their Gerber preview before ordering